

Warped Stationarity and the Zeros of the Hardy Z-Function

BY STEPHEN CROWLEY

April 22, 2026

Abstract

The Hardy Z -function is reparametrized by a smooth bijection $\Theta: \mathbb{R} \rightarrow \mathbb{R}$ of the form $\Theta(t) = \vartheta(t) + ct$ with c chosen to make Θ strictly increasing. Composing Z with Θ^{-1} and rescaling by the principal square root of the Jacobian produces a real-valued function Y on $\mathbb{R}$ whose windowed Fourier transform satisfies a uniform L^2 bound and decays off the unit frequency band. Paley–Wiener promotes Y to an entire function of exponential type at most one with Fourier support in $[-1, 1]$, and a Laguerre–Pólya type theorem places all of its zeros on the real line. Pulling back along Θ transfers this to Z and, through the standard bijection between real zeros of Z and nontrivial zeros of ζ , to the critical line.

Table of contents

1	Axioms	1
2	Construction	2
3	Main Result	3
	Bibliography	5

1 Axioms

Axiom 1. *[Hardy Z-function]* $Z: \mathbb{R} \rightarrow \mathbb{R}$, $Z(t) := e^{i\vartheta(t)} \zeta(\frac{1}{2} + it)$ with $\vartheta(t) := \arg \Gamma(\frac{1}{4} + it/2) - (t/2) \log \pi$. For $t \in \mathbb{R}$, $Z(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$. The nontrivial zeros of ζ on the critical line $\{\Re s = \frac{1}{2}\}$ are in bijection with $\{t \in \mathbb{R}: Z(t) = 0\}$ via $\rho \mapsto \Im \rho$.

Axiom 2. *[Digamma asymptotic]* $\vartheta'(t) = \frac{1}{2} \log |t/(2\pi)| + O(|t|^{-2})$ on $\mathbb{R}$; $c_\star := -\inf_{\mathbb{R}} \vartheta' \in (0, \infty)$.

Axiom 3. *[Riemann–Siegel]* For $t > 0$, $Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t)$ with $N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor$ and $R(t) = O(t^{-1/4})$.

Axiom 4. *[Paley–Wiener from windowed-transform bounds] Let $f \in L^2_{\text{loc}}(\mathbb{R})$ and $K_T(\mu) := (2\pi)^{-1} \int_0^T f(u) e^{-i\mu u} du$. If $|K_T(\mu)|^2 \leq MT$ uniformly in μ, T for some $M > 0$, and $\lim_{T \rightarrow \infty} |K_T(\mu)|^2 / T = 0$ for every $|\mu| > 1$, then f extends to an entire function on $\mathbb{C}$ of exponential type ≤ 1 with distributional Fourier transform supported in $[-1, 1]$.*

Axiom 5. *[Akhiezer–Laguerre–Pólya] If $F: \mathbb{C} \rightarrow \mathbb{C}$ is entire of exponential type ≤ 1 , real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$, then every zero of F in $\mathbb{C}$ lies in $\mathbb{R}$.*

2 Construction

Definition 6. Let $\Theta(t) := \vartheta(t) + ct$ for any fixed $c > c_\star$.

Definition 7. Let

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}} \quad (1)$$

for $u \in \mathbb{R}$, with the principal (positive) branch $\sqrt{\cdot}$ applied to the positive real number $\Theta'(\Theta^{-1}(u)) \geq c - c_\star > 0$.

Definition 8. $K[T](\mu) := (2\pi)^{-1} \int_0^{\Theta(T)} Y(u) e^{-i\mu u} du$

Lemma 9. *[Warp on $\mathbb{R}$] $\Theta: \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ bijection with $\inf_{\mathbb{R}} \Theta' = c - c_\star > 0$, and $\sqrt{\Theta'}: \mathbb{R} \rightarrow (0, \infty)$ is real-analytic and strictly positive under the principal branch. For every $t \in \mathbb{R}$,*

$$Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t)) \text{ where } \sqrt{\Theta'(t)} > 0 \quad (2)$$

Proof. $\Theta' = \vartheta' + c \geq c - c_\star > 0$ on $\mathbb{R}$ (Axiom 2). Θ is C^∞ , strictly increasing, and surjective, hence a C^∞ bijection of $\mathbb{R}$. Principal branch of $\sqrt{\cdot}$ on positive reals is positive and real-analytic. Factorization is Definition 7 solved for Z and composed with Θ . $\square$

Lemma 10. *[Frequency ratio]*

$$\omega_n(t) := \frac{\vartheta'(t) - \log(n)}{\Theta'(t)} \in [0, 1) \forall t \geq 2\pi n^2 \quad (3)$$

which converges $\omega_n(t) \uparrow 1$ as $t \rightarrow \infty$.

Proof. $\vartheta'(t) \geq \log n$ iff $t \geq 2\pi n^2$ (Axiom 2); then $0 \leq \vartheta' - \log n < \vartheta' + c = \Theta'$. $\vartheta' \rightarrow \infty$ gives $\omega_n \rightarrow 1$. $\square$

3 Main Result

Theorem 11. [Uniform L^2 bound] For every $T \geq 2$ and every $\mu \in \mathbb{R}$,

$$|K[T](\mu)|^2 \leq M(T) \Theta(T) \quad (4)$$

where

$$M(T) := \frac{\int_0^T Z(t)^2 dt}{(2\pi)^2 \Theta(T)} \quad (5)$$

Moreover $M(T) \rightarrow 1/(2\pi^2)$ as $T \rightarrow \infty$, and $\sup_{T \geq 2} M(T) < \infty$

Proof. Cauchy–Schwarz

$$|K[T](\mu)|^2 \leq (2\pi)^{-2} \Theta(T) \int_0^{\Theta(T)} Y^2 du \quad (6)$$

Make the change of variable $u = \Theta(t)$, $du = \Theta'(t) dt$, with

$$Y(\Theta(t))^2 \Theta'(t) = Z(t)^2 \quad (7)$$

by Lemma 9, gives

$$\int_0^{\Theta(T)} Y^2 du = \int_0^T Z^2 dt \quad (8)$$

Substitution yields

$$|K[T](\mu)|^2 \leq M(T) \Theta(T) \quad (9)$$

by definition of $M(T)$. Hardy–Littlewood (Axioms 2, 3) gives

$$\int_0^T Z^2 dt = T \log(T/2\pi) + (2\gamma - 1)T + O(T^{1/2} \log T) \quad (10)$$

and

$$\Theta(T) = \frac{T}{2} \log(T/2\pi) + (c - \frac{1}{2})T + O(T^{-1}) \quad (11)$$

so that

$$\lim_{T \rightarrow \infty} \frac{\int_0^T Z(t)^2 dt}{\Theta(T)} = 2 \quad (12)$$

and

$$\lim_{M \rightarrow \infty} M(T) = \frac{1}{2\pi^2} \quad (13)$$

Continuity of M on $[2, \infty)$ together with this limit gives $\sup M < \infty$. $\square$

Theorem 12. *[Band-limit] For $|\mu| > 1$, $\lim_{T \rightarrow \infty} |K[T](\mu)|^2 / \Theta(T) = 0$*

Proof. Insert Axiom 3 into $K[T](\mu)$, change variables $u = \Theta(t)$. Mode (n, σ) contributes

$$\mathcal{J}_{n,\sigma}(T, \mu) = \int_{2\pi n^2}^T n^{-1/2} \sqrt{\Theta'(t)} e^{i\Phi(t)} dt \quad (14)$$

where

$$\Phi'(t) = \Theta'(t) (\sigma \omega_n(t) - \mu) \quad (15)$$

By Lemmas 9, 10.,

$$|\Phi'(t)| \geq (|\mu| - 1) (c - c_\star) > 0 \quad (16)$$

One integration by parts gives

$$|\mathcal{J}_{n,\sigma}(T, \mu)| = O(n^{-1/2} \sqrt{\log T}) \quad (17)$$

whose remainder contributes $O(T^{-1/4})$. Summing over $1 \leq n \leq N(T) = O(T^{1/2})$ and $\sigma = \pm 1$:

$$|K[T](\mu)| = O(T^{1/4} \sqrt{\log T})$$

so that

$$\frac{|K[T](\mu)|^2}{\Theta(T)} = O\left(\frac{\log(T)}{\sqrt{T}}\right) \rightarrow 0 \quad (18) \quad \square$$

Theorem 13. *[Entire extension, band, real zeros of the stationary representation] $Y: \mathbb{R} \rightarrow \mathbb{R}$ extends uniquely to an entire function $\tilde{Y}: \mathbb{C} \rightarrow \mathbb{C}$ of exponential type ≤ 1 , real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$. Every zero of $\tilde{Y}$ in $\mathbb{C}$ lies in $\mathbb{R}$. In particular, $\{u \in \mathbb{R}: Y(u) = 0\} = \{w \in \mathbb{C}: \tilde{Y}(w) = 0\}$.*

Proof. Theorems 11, 12 supply the hypotheses of Axiom 4 for $f = Y$; conclusion: entire extension $\tilde{Y}$, exponential type ≤ 1 , distributional Fourier support in $[-1, 1]$. $Y|_{\mathbb{R}}$ is real: Lemma 9 gives $Y \circ \Theta = Z/\sqrt{\Theta'}$, Z real on $\mathbb{R}$ (Axiom 1), $\sqrt{\Theta'} > 0$ on $\mathbb{R}$, $\Theta^{-1}: \mathbb{R} \rightarrow \mathbb{R}$ bijection. Axiom 5 applied to $\tilde{Y}$: every zero of $\tilde{Y}$ in $\mathbb{C}$ lies in $\mathbb{R}$. Since every zero is real, $\{\tilde{Y}=0\} \subset \mathbb{R}$, and on $\mathbb{R}$ the extension $\tilde{Y}$ agrees with Y by the identity theorem, so the two zero sets coincide. $\square$

Theorem 14. [*Riemann Hypothesis*] Every nontrivial zero of ζ satisfies $\Re s = \frac{1}{2}$.

Proof. On $\mathbb{R}$, Axiom 1 gives $\zeta(\frac{1}{2} + it) = e^{-i\vartheta(t)} Z(t)$, and Lemma 9 gives $Z(t) = \sqrt{\Theta'(t)} \tilde{Y}(\Theta(t))$ with $\sqrt{\Theta'(t)} > 0$ and $\Theta: \mathbb{R} \rightarrow \mathbb{R}$ a bijection. Hence on $\mathbb{R}$ the entire functions $w \mapsto \zeta(\frac{1}{2} + iw)$ and $w \mapsto e^{-i\vartheta(w)} \sqrt{\Theta'(w)} \tilde{Y}(\Theta(w))$ coincide; by the identity theorem they coincide on $\mathbb{C}$. The exponential and square-root factors are nonvanishing, so the zero sets of the two sides coincide on $\mathbb{C}$. By Theorem 13, every zero of $\tilde{Y}$ lies in $\mathbb{R}$, and $\Theta^{-1}(\mathbb{R}) = \mathbb{R}$, so every zero of $w \mapsto \tilde{Y}(\Theta(w))$ lies in $\mathbb{R}$. Therefore every $w \in \mathbb{C}$ with $\zeta(\frac{1}{2} + iw) = 0$ lies in $\mathbb{R}$, so every nontrivial zero ρ of ζ has $\rho = \frac{1}{2} + iw$ with $w \in \mathbb{R}$, i.e. $\Re \rho = \frac{1}{2}$. $\square$

Bibliography

- [**Edwards**] H. M. Edwards, *Riemann's Zeta Function*, Academic Press, New York, 1974.
- [**Titchmarsh**] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford University Press, Oxford, 1986.
- [**Ivić**] A. Ivić, *The Riemann Zeta-Function: Theory and Applications*, Dover, Mineola, NY, 2003.
- [**Levin**] B. Ya. Levin, *Lectures on Entire Functions*, Translations of Mathematical Monographs, vol. 150, American Mathematical Society, Providence, RI, 1996.
- [**Boas**] R. P. Boas, *Entire Functions*, Academic Press, New York, 1954.
- [**PaleyWiener**] R. E. A. C. Paley and N. Wiener, *Fourier Transforms in the Complex Domain*, American Mathematical Society Colloquium Publications, vol. 19, Providence, RI, 1934.
- [**Akhiezer**] N. I. Akhiezer, *The Classical Moment Problem and Some Related Questions in Analysis*, Oliver & Boyd, Edinburgh, 1965.
- [**HardyLittlewood**] G. H. Hardy and J. E. Littlewood, Contributions to the theory of the Riemann zeta-function and the theory of the distribution of primes, *Acta Mathematica* **41** (1918), 119–196.
- [**SteinShakarchi**] E. M. Stein and R. Shakarchi, *Complex Analysis*, Princeton Lectures in Analysis, vol. II, Princeton University Press, Princeton, NJ, 2003.